

Discrete Fourier Analysis

2.1.3 Discrete Fourier analysis in detail

2.1.3 Discrete Fourier Analysis (Detailed Explanation)

English + Nepali Mixup (No Romanize)

Introduction

Discrete Fourier Analysis (DFA) भनेको **Discrete-Time Signal** लाई **frequency components** मा divide गरेर analysis गर्ने mathematical technique हो।

यसले signal भित्र कुन-कुन frequency हरू छन्, तिनीहरूको magnitude (strength) र phase (position) कति छ भन्ने जानकारी दिन्छ।

Simple शब्दमा,

Time Domain → **Frequency Domain** मा signal लाई convert गर्ने process नै **Discrete Fourier Analysis** हो।

Real Life Idea

Imagine गर्नुहोस् एउटा music song छ।

हामीले सुन्दा एउटा मात्र sound जस्तो लाग्छ।

तर वास्तवमा त्यो song धेरै frequencies मिलेर बनेको हुन्छ।

जस्तै

- Drum → Low Frequency
- Guitar → Medium Frequency
- Flute → High Frequency
- Vocal → Different Frequency Combination

Discrete Fourier Analysis ले यी सबै frequencies छुट्टाछुट्टै निकालेर देखाउँछ।

Why Discrete Fourier Analysis is Needed?

Real world signals rarely contain only one frequency.

Example

Voice Signal

यसमा हुन्छ

- Fundamental frequency
- Harmonics
- Noise

यदि signal लाई frequency मा analyze नगर्ने हो भने

- Noise हटाउन सकिँदैन
- Compression गर्न सकिँदैन
- Equalizer बनाउन सकिँदैन
- Signal detect गर्न सकिँदैन

त्यसैले Fourier Analysis आवश्यक हुन्छ।

Domains of Signal

Signal लाई दुई तरिकाले represent गर्न सकिन्छ।

1. Time Domain

यसमा signal amplitude समय अनुसार change हुन्छ।

Example

Time
0 1 2 3 4 5
Amplitude
2 5 7 3 1 0

हामी amplitude मात्र देखौं।

Frequency देखिँदैन।

2. Frequency Domain

यसमा signal कुन frequency बाट बनेको छ भन्ने देखिन्छ।

Example

50 Hz
100 Hz
200 Hz
400 Hz

प्रत्येक frequency को magnitude अलग हुन्छ।

Basic Concept

Suppose एउटा signal छ

$x[n]$

यसलाई Fourier Analysis गरेपछि

$X[k]$

मिल्छ।

जहाँ

$x[n]$

↓

Fourier Analysis

↓

$X[k]$

यहाँ

$x[n]$ = Time Domain

$X[k]$ = Frequency Domain

What is Frequency?

Frequency भन्नाले

एक second मा wave कति पटक repeat हुन्छ भन्ने हो।

Unit

Hertz (Hz)

Example

100 Hz

=

100 cycles/sec

What is Fourier Transform?

Fourier Transform ले signal लाई धेरै sine र cosine waves मा divide गर्छ।

Example

Original Signal

=

Low Frequency

+

Medium Frequency

+

High Frequency

Discrete Fourier Transform (DFT)

Discrete signal को Fourier Analysis गर्न प्रयोग हुने method लाई

Discrete Fourier Transform (DFT) भनिन्छ।

Formula

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}$$

जहाँ

- **x(n)** = Input Signal
- **X(k)** = Frequency Component
- **N** = Total Samples
- **k** = Frequency Index
- **j** = Imaginary Number ($\sqrt{-1}$)

Understanding the Formula

Suppose

N = 8

अब DFT ले

8 samples

↓

8 frequencies

produce गर्छ।

Example

Sample

↓

Frequency
0
↓
0 Hz
1
↓
Low Frequency
2
↓
Medium Frequency
3
↓
Higher Frequency
...
7

Output of DFT

DFT ले दुईवटा information दिन्छ।

1. Magnitude

बताउँछ

Frequency कति strong छ।

Example

Frequency
100 Hz
Magnitude
20

मतलब

100 Hz धेरै strong छ।

2. Phase

बताउँछ

Wave कहाँबाट start भएको हो।

Unit

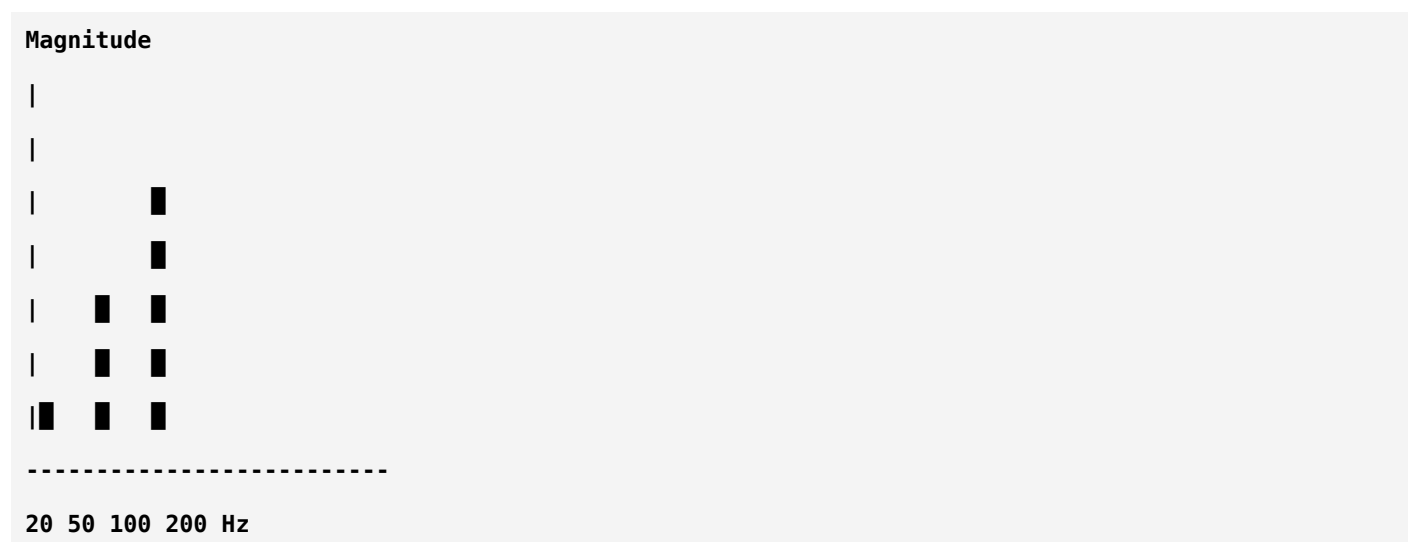
Degree
or
Radian

Magnitude Spectrum

Magnitude Spectrum भनेको

सबै frequencies को strength देखाउने graph हो।

Example



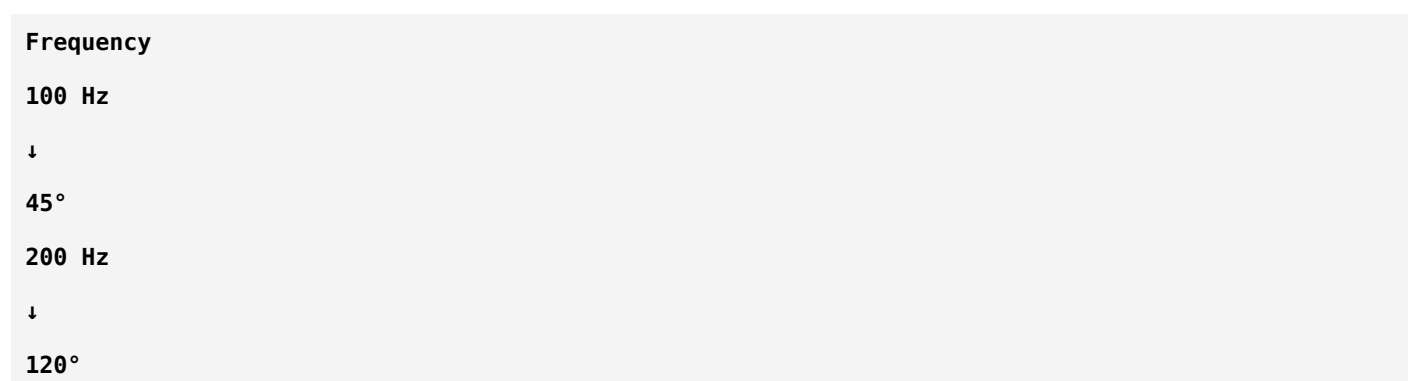
100 Hz सबैभन्दा strong छ।

Phase Spectrum

Phase Spectrum ले

प्रत्येक frequency को phase देखाउँछ।

Example



Sampling Relation

Suppose

Sampling Frequency

$F_s = 1000 \text{ Hz}$

त्यसपछि

Maximum Detectable Frequency

$F_s/2$

=

500 Hz

यसलाई Nyquist Frequency भनिन्छ।

Frequency Resolution

Frequency Resolution

बताउँछ

दुई frequency बीच minimum difference।

Formula

$$\Delta f = \frac{F_s}{N}$$

Example

$F_s = 1000 \text{ Hz}$

$N = 100$

Resolution

=

10 Hz

मतलब

0

10

20

30

40

...

1000

Example

Suppose

Signal

```
x[n]
=
sin(50 Hz)
+
sin(150 Hz)
```

Time domain मा हेर्दा

Complex waveform देखिन्छ।

Fourier Analysis गरेपछि

```
50 Hz
↓
Peak
150 Hz
↓
Peak
```

दुईटा frequency मात्र देखिन्छ।

Steps of Discrete Fourier Analysis

Step 1

Signal Collect गर्नुहोस्।

```
Microphone
↓
Digital Signal
```

Step 2

Sampling गर्नुहोस्।

Continuous

↓

Discrete

Step 3

DFT Apply गर्नुहोस्।

Time Domain

↓

Frequency Domain

Step 4

Magnitude निकाल्नुहोस्।

Step 5

Phase निकाल्नुहोस्।

Step 6

Frequency Analyze गर्नुहोस्।

Fast Fourier Transform (FFT)

DFT calculate गर्न धेरै time लाग्छ।

त्यसैले

Fast Fourier Transform (FFT)

प्रयोग गरिन्छ।

FFT

DFT calculate गर्ने efficient algorithm हो।

Comparison

DFT	FFT
Slow	Very Fast
High Computation	Low Computation
$O(N^2)$	$O(N \log_2 N)$
Large Signal मा Slow	Large Signal मा Fast

Applications

1. Audio Processing

- Noise Removal
- Music Analysis
- Voice Recognition

2. Image Processing

- Image Compression
 - Edge Detection
 - Image Enhancement
-

3. Communication System

- OFDM
 - Spectrum Analysis
 - Channel Estimation
-

4. Medical Field

- ECG Analysis
 - EEG Analysis
 - MRI Signal Processing
-

5. Radar System

- Object Detection
 - Speed Measurement
-

6. Speech Processing

- Voice Recognition
 - Speaker Identification
-

7. Wireless Communication

- Wi-Fi
 - 4G
 - 5G
-

8. Seismology

- Earthquake Analysis
- Vibration Analysis

Advantages

- ✓ Time domain signal लाई frequency domain मा convert गर्छ।
- ✓ Hidden frequencies सजिलै पत्ता लगाउन सकिन्छ।
- ✓ Noise identify र remove गर्न मद्दत गर्छ।
- ✓ Compression algorithms मा उपयोगी हुन्छ।
- ✓ Audio, image, speech, radar, medical लगायत धेरै क्षेत्रमा प्रयोग हुन्छ।
- ✓ FFT प्रयोग गर्दा computation धेरै छिटो हुन्छ।

Disadvantages

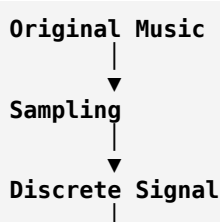
- ✗ DFT को computation cost धेरै हुन्छ।
- ✗ Long signals का लागि processing time बढी लाग्न सक्छ।
- ✗ Frequency resolution sample size (**N**) मा निर्भर हुन्छ।
- ✗ Non-stationary signals का लागि साधारण DFT पर्याप्त नहुन सक्छ; त्यसका लागि **Short-Time Fourier Transform (STFT)** वा **Wavelet Transform** प्रयोग गरिन्छ।

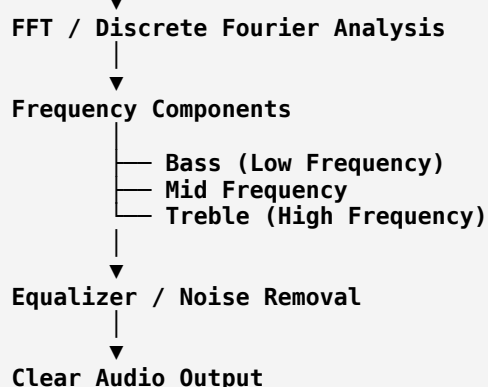
Real-Life Example (Music Player 🎵)

Imagine गर्नुहोस् तपाईंले एउटा **MP3 song** play गर्नुभयो।

1. Song microphone बाट record हुन्छ।
2. Analog signal लाई **sampling** गरेर digital signal बनाइन्छ।
3. **Discrete Fourier Analysis (FFT)** प्रयोग गरेर frequency components निकालिन्छ।
4. Equalizer ले Bass, Mid, Treble जस्ता specific frequencies लाई increase वा decrease गर्छ।
5. Noise reduction ले unwanted frequencies हटाउँछ।
6. अन्त्यमा तपाईंले clear र high-quality sound सुन्नुहुन्छ।

Flow:





Summary

Discrete Fourier Analysis भनेको **Discrete-Time Signal** लाई **Frequency Domain** मा analyze गर्ने technique हो। यसको मुख्य उद्देश्य signal भित्र रहेका विभिन्न frequency components, तिनको **magnitude** र **phase** पत्ता लगाउनु हो। यसको लागि **Discrete Fourier Transform (DFT)** प्रयोग गरिन्छ, र व्यवहारिक रूपमा computation छिटो बनाउन **Fast Fourier Transform (FFT)** algorithm प्रयोग गरिन्छ। Audio processing, image processing, communication systems, medical signal analysis, radar, speech recognition लगायतका क्षेत्रमा यसको अत्यधिक प्रयोग हुन्छ।

image illustration also

2.1.3 DISCRETE FOURIER ANALYSIS (DFA)

Discrete Fourier Analysis (DFA) transforms a discrete-time signal from the **time domain** to the **frequency domain** to find the frequency components present in the signal.

